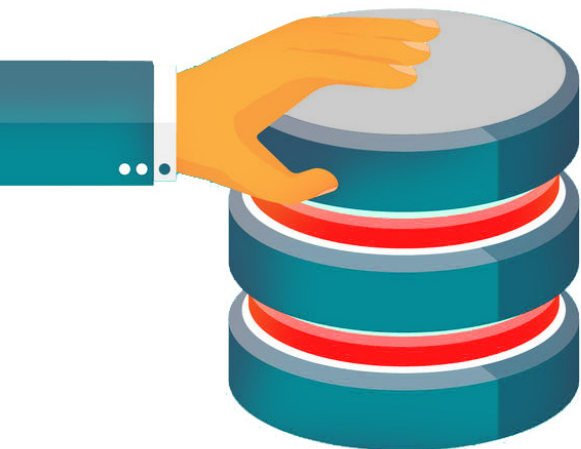


PostgreSQL: An Advanced and Versatile Open Source RDBMS

‘Data is the new oil’ is passé. The most recent cliché is ‘Data is water’. More and more organisations are in a race against time to extract insights from their data to provide better value propositions to their customers so as to increase stakeholder value. It thus becomes imperative to choose the right OLTP and OLAP database and storage to enable faster time to market for the developer community. This article focuses on five major advantages of PostgreSQL.



Incubated in the University of California just about three decades ago, PostgreSQL, the open source relational database management system, has become a household name among database developers and administrators. The name Postgres comes from the project name under which it was built — Ingres DBMS (Ingres being an abbreviation for Interactive Graphics REtrieval System). It was eventually renamed Postgres. Postgres is mostly qualified as an object relational database management system (ORDBMS) comprising a relational database (RDBMS) and an object-oriented database (OODBMS) allowing inheritance. ORDBMS has a feature that allows developers to build and innovate their own data types and methods, which can be applied to the DBMS. The intention is to allow developers to increase the abstraction with which they view the problem area.

Advantage 1: Three decades of popularity, illustrating the steadfastness of the platform

A quick look at the evolution of PostgreSQL provides several insights into the future of this formidable open source platform, namely:

- A strong roadmap shows stability of the platform
 - Rapid innovation in this platform is proof of its adoption
 - Its wide range of features showcase its versatility
- Some of the notable features include:
- A benchmarking application to carry out various types of performance tests against PostgreSQL.

In version 7.4:

- A simple utility program removes any ‘orphaned’ large objects from a PostgreSQL database released as part of version 8.3. This version upwards also provides various cryptographic functions.
- A library to support regression testing of the SECURITY LABEL statement as part of version 9.1.

With the coming of Version 14, which is available for download, one can look out for advanced features such as:

- Maximum parallel worker increase
- Advanced security features
- Performance insights
- Memory lock-ins

Advantage 2: Continuous feature addition, exhibiting innovation at the core

The rapid growth and stability of this platform has made it not just the favourite community for starter kits but also a popular choice in enterprise scale solutions. The versatility of this RDBMS is incomparable in the open source space. PostgreSQL is versatile not only because of its ability to store structured and semi-structured data such as XML and JSON, but also because it provides an ideal platform for OLTP and OLAP workloads. It provides traditional row